Endpoint Imaging, Inventory & Hardware Lifecycle Management

Implementation Solution Guide

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Executive Summary — Endpoint Imaging, Inventory & Hardware Lifecycle Management

Without a defined lifecycle, devices appear and vanish off the books: nobody knows what was bought, what is deployed, what is idle in a drawer, or what was retired without being wiped. Imaging is done ad hoc, inventory lives in a stale spreadsheet, and audits are painful. This package gives you a repeatable endpoint imaging, inventory, and hardware lifecycle — a clear provisioning method (cloud provisioning with Windows Autopilot, or Configuration Manager task-sequence imaging where it is genuinely required), accurate hardware inventory as a source of truth, and a governed procure → deploy → refresh → retire flow that ends every device's life with a secure wipe.

This is a complete, deploy-it-today package, written for an administrator with no prior device-lifecycle experience. It tells you which provisioning method to pick and why, how to make Intune the system of record for hardware inventory, and exactly how to retire and wipe a device so it leaves the fleet clean.

What you get

- A plain-language Solution Overview of the device lifecycle model, the provisioning decision

(Autopilot vs. imaging), hardware inventory sources, and the procure/deploy/refresh/retire flow.

- A Requirements Guide with verify / implement-if-missing for every prerequisite.
- An Implementation Guide of precise, click-by-click steps in the required order.
- Five PowerShell scripts for hardware/inventory discovery, provisioning and inventory validation,

a lifecycle and asset-reconciliation report, a safe retire/wipe remediation helper (preview by default), and a lifecycle-stage group builder.

- Testing and Troubleshooting guides for a confident rollout.

What this delivers

A defined lifecycle for every device: one chosen provisioning method, an accurate hardware inventory you can trust for audits and refresh planning, lifecycle-stage groups that move a device cleanly from procured to deployed to refresh-due to retired, and a secure wipe at end of life.

The repeatable deployment model

Assess → Configure → Pilot → Validate → Expand.

Outcome

Every device accounted for across its whole life — provisioned by a defined method, inventoried accurately, refreshed on criteria, and retired with a secure wipe — proven on a pilot before you scale.

1. Solution Overview — Endpoint Imaging, Inventory & Hardware Lifecycle Management

What this solution is

A repeatable way to manage the whole life of every Windows device: how it gets provisioned, how it is inventoried, when it gets refreshed, and how it is securely retired. It replaces ad hoc imaging and a stale spreadsheet with a defined provisioning method, an accurate hardware inventory that acts as the source of truth, and a governed procure → deploy → refresh → retire flow that ends in a secure wipe.

The device lifecycle model

1. Procure — a device is purchased and its hardware identity (hardware hash or serial / manufacturer / model) is captured before or at receipt.

1. Deploy — the device is provisioned by your chosen method (Windows Autopilot cloud provisioning or Configuration Manager task-sequence imaging) and enrolled in Intune.

1. Operate / inventory — the device is in service and its hardware details are tracked in Intune, which is the inventory system of record.

1. Refresh — when the device meets refresh criteria (age, warranty expiry, or spec), it is flagged for replacement and its successor is procured.

1. Retire — at end of life the device is retired and securely wiped, then removed from inventory (and de-registered from Autopilot if applicable).

The provisioning decision (Autopilot vs. imaging)

This is the most important choice in the solution, so make it deliberately.

• Windows Autopilot cloud provisioning is the default for modern, cloud-managed fleets. The device ships from the vendor (or is reset), the user signs in, and Intune provisions it over the internet with no custom image. You register the device's hardware identity once; everything else is policy. Choose this when devices are internet-connected, you want zero-touch deployment, and you do not need a custom on-disk image.

• Configuration Manager task-sequence imaging is the alternative for environments that genuinely require it: an on-premises image with a specific disk layout or drivers, offline or bandwidth-constrained sites, or a managed estate already standardized on Configuration Manager. It lays down an OS image and applications from a distribution point on the LAN.

Why the choice matters: it decides where your "golden state" lives (Microsoft cloud policy vs. an on-prem image), how new hardware is onboarded, and how much network you need at deploy time. Most fleets should standardize on Autopilot and keep imaging only for the sites or device types that truly need it.

Hardware inventory as a source of truth

Intune holds per-device hardware details (manufacturer, model, serial number, processor, memory, storage, enrollment and last-check-in dates). Once devices are enrolled, this becomes the single inventory you reconcile against — not a spreadsheet. Autopilot adds a registered-devices list keyed by hardware

hash, which is your procurement-to-deployment ledger. Reconciling purchased devices against enrolled devices is how you find machines that were bought but never deployed, or deployed but never inventoried.

The procure → deploy → refresh → retire flow with secure wipe

Lifecycle-stage security groups move a device through the flow and let policy and reporting target the right stage. The solution names them with a fixed convention so they are reusable verbatim:

- `Device Lifecycle – Procurement – Procured`
- `Device Lifecycle – Fleet – Deployed`
- `Device Lifecycle – Aging – Refresh-Due`
- `Device Lifecycle – End-of-Life – Retire`

At retirement, Intune performs a secure wipe. Retire removes company data and management; Wipe returns the device to a clean factory state (a full reset). Choosing the correct action at end of life is what guarantees no data leaves with a retired device.

Benefits

- Accountability — every device is tracked from procurement to retirement; nothing falls off the books.
- A defined provisioning method — one repeatable way to stand up a device, chosen deliberately.
- Accurate inventory — Intune is the source of truth for audits and refresh planning, not a spreadsheet.
- Clean retirement — every retired device is securely wiped, so no data walks out the door.
- Proven before scale — piloted on a small group before the whole fleet.

Limitations (be honest with buyers)

- Autopilot needs the hardware identity registered (hash or vendor registration) and internet connectivity at provisioning; it does not lay down a custom on-disk image.
- Task-sequence imaging requires a working Configuration Manager environment (site, distribution point, boot media); standing that up is out of scope here, though we note where it fits.
- Intune hardware inventory reflects enrolled devices; devices bought but never enrolled appear in the reconciliation as gaps, not in inventory.
- A secure wipe is irreversible — confirm the device and the action before applying.

The repeatable deployment model

> *Choose the provisioning method → register/import devices and establish inventory → build the lifecycle-stage groups → pilot the flow → validate → expand to the fleet. The repeatable way: Assess → Configure → Pilot → Validate → Expand.*

2. Requirements Guide — Endpoint Imaging, Inventory & Hardware Lifecycle Management

Verify every requirement before configuring. Each entry: Description · Why · Verify · Implement-if-missing. Category-wide prerequisites are in `global-requirements-infrastructure.md`.

2.1 Intune (provisioning, inventory, and retire/wipe)

Description: Microsoft Intune (Plan 1), the control plane for Autopilot, enrollment, hardware inventory, and the retire/wipe actions. Why: Autopilot profiles, device inventory, and the secure-wipe lifecycle actions all live in Intune. Verify: `intune.microsoft.com` loads; Devices is visible. Implement: Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

2.2 Provisioning method chosen (Autopilot or imaging)

Description: A deliberate decision: Windows Autopilot cloud provisioning (default) or Configuration Manager task-sequence imaging (where required). Why: This decides where the golden state lives and how devices are onboarded; the implementation steps differ. Verify: Confirm which method this fleet (or this site) will use; run `IMGLIFE-Discovery.ps1` to see current state (Autopilot registration, enrollment, any ConfigMgr client). Implement: Default to Autopilot. Use Configuration Manager imaging only for sites/device types that require a custom on-disk image or offline deployment, and confirm a working ConfigMgr site exists.

2.3 Windows Autopilot prerequisites (if using Autopilot)

Description: Devices' hardware identities registered to Autopilot (vendor registration or imported hardware hash) and an Autopilot deployment profile. Why: Autopilot can only provision devices it recognizes by hardware identity. Verify: Intune → Devices → Enrollment → Windows Autopilot → Devices shows the registered devices; run `IMGLIFE-Validation.ps1`. Implement: Have the OEM/reseller register devices, or import the hardware hash; create an Autopilot deployment profile (Step 2 of the Implementation Guide).

2.4 Configuration Manager environment (if using imaging)

Description: A working ConfigMgr site with a distribution point, boot media, an OS image package, and a task sequence. Why: Task-sequence imaging deploys the OS and apps from on-premises distribution points. Verify: ConfigMgr console → Software Library → Operating Systems → Task Sequences shows a usable deployment task sequence. Implement: Stand up (or reuse) a ConfigMgr site and build an OS-deployment task sequence; this is out of scope here but noted as the imaging path.

2.5 Device enrollment in Intune

Description: Provisioned devices are enrolled and manageable in Intune. Why: Inventory, lifecycle groups, and retire/wipe only reach enrolled devices. Verify: Intune → Devices → All devices shows the device; run `IMGLIFE-Validation.ps1`. Implement: Enable automatic enrollment and join devices to Entra ID (Autopilot enrolls automatically; imaged devices enroll via the join/enrollment step).

2.6 Hardware inventory source of truth

Description: Intune device hardware inventory is treated as the authoritative inventory, with a procurement record to reconcile against. Why: An accurate, single source of truth is what makes audits, refresh planning, and reconciliation possible. Verify: Intune → Devices → All devices → (a device) → Hardware shows model, serial, storage; run `IMGLIFE-Report.ps1` for the fleet inventory. Implement: Ensure devices are enrolled (so they report hardware); keep a procurement list (serials) to reconcile against the enrolled inventory.

2.7 Retirement and secure-wipe process

Description: A defined end-of-life action — Retire (remove company data/management) or Wipe (full factory reset) — and who is authorized to run it. Why: A device must leave the fleet clean; an undocumented retirement is how data walks out the door. Verify: Confirm the operator holds Intune Administrator and that the action is documented; preview with `IMGLIFE-Remediation.ps1` (runs in WhatIf mode by default). Implement: Document the retire-vs-wipe decision and authorization; use `IMGLIFE-Remediation.ps1` as the safe helper.

2.8 Lifecycle-stage groups

Description: Entra ID security groups for each lifecycle stage, named by the solution convention. Why: Stages let policy and reporting target the right population as a device moves through its life. Verify: Entra → Groups shows the four lifecycle groups; or run `IMGLIFE-Set-LifecycleStageTargeting.ps1` (idempotent — reports/creates). Implement: Run `IMGLIFE-Set-LifecycleStageTargeting.ps1` to create them, or create them by hand with the exact names below.

Checklist

- Intune licensing assigned
- Provisioning method chosen (Autopilot default, or ConfigMgr imaging) — `IMGLIFE-Discovery.ps1`
- Autopilot devices registered + deployment profile (if Autopilot) — `IMGLIFE-Validation.ps1`
- ConfigMgr site + task sequence (if imaging)
- Devices enrolled in Intune
- Hardware inventory established as source of truth — `IMGLIFE-Report.ps1`
- Retire/wipe action documented and authorized
- Lifecycle-stage groups created — `IMGLIFE-Set-LifecycleStageTargeting.ps1`

3. Implementation Guide — Endpoint Imaging, Inventory & Hardware Lifecycle Management

Follow these steps in order, in the Microsoft Intune admin center (`https://intune.microsoft.com`). The default path is Windows Autopilot cloud provisioning; Step 3 covers the Configuration Manager task-sequence imaging alternative for sites that require it.

Model: choose one provisioning method, register/import devices and establish inventory, build the lifecycle-stage groups, then pilot the procure → deploy → refresh → retire flow before expanding.

Step 1 — Create the lifecycle-stage groups

Create these four Entra ID security groups (Entra → Groups → New group → Security), using the exact names so the rest of the solution can reuse them verbatim:

1. `Device Lifecycle – Procurement – Procured`
2. `Device Lifecycle – Fleet – Deployed`
3. `Device Lifecycle – Aging – Refresh-Due`
4. `Device Lifecycle – End-of-Life – Retire`

Also create the pilot provisioning group `Autopilot Provisioning – Standard Users – Pilot` (a few representative users/devices) and `Autopilot Provisioning – Standard Users – Broad` (the rest). Tip: `IMGLIFE-Set-LifecycleStageTargeting.ps1` creates these idempotently.

Step 2 — Register devices and create the Autopilot deployment profile (default path)

1. Register hardware identities: Devices → Enrollment → Windows Autopilot → Devices. Devices registered by your OEM/reseller appear here; otherwise import the hardware hash (CSV) with Import.
1. Create the profile: Devices → Enrollment → Windows Autopilot → Deployment profiles → Create profile → Windows PC.
1. Name: `Autopilot Provisioning – Standard Users – Pilot`. Next.
2. Out-of-box experience (OOBE): set deployment mode (User-driven), join type (Microsoft Entra joined), and the OOBE options (hide EULA, hide privacy settings, etc.) to your policy. Next.
1. Assignments → Add groups → `Autopilot Provisioning – Standard Users – Pilot`. Next → Create.
2. Repeat for `Autopilot Provisioning – Standard Users – Broad` assigned to the broad group (do this after the pilot validates).

Step 3 — Alternative: Configuration Manager task-sequence imaging (only if required)

Use this instead of Step 2 only for sites/device types that need a custom on-disk image or offline deployment. In the Configuration Manager console:

1. Software Library → Operating Systems → confirm an OS image package and a deployment task sequence exist on a distribution point.
1. Deploy the task sequence to the imaging collection and boot the device from PXE or boot media.

2. After imaging, ensure the device enrolls into Intune (Entra join + automatic enrollment) so it reports hardware inventory like any other device.

Step 4 — Establish hardware inventory as the source of truth

1. Confirm enrolled devices report hardware: Devices → All devices → (a device) → Hardware (model, serial number, storage, memory).

1. Pull the fleet inventory with `IMGLIFE-Report.ps1` (Microsoft Graph), which lists managed devices plus the Autopilot registered-devices list.

1. Reconcile your procurement list (serials purchased) against the report to find gaps: bought but not enrolled, or enrolled but not on the books.

Step 5 — Move devices into the Deployed stage

1. As devices finish provisioning and report inventory, add them to `Device Lifecycle – Fleet – Deployed`.

1. Keep `Device Lifecycle – Procurement – Procured` for devices received but not yet provisioned, so the reconciliation report can flag procured-but-not-deployed machines.

Step 6 — Define refresh criteria and flag refresh-due devices

1. Decide the refresh trigger (for example, age over N years, warranty expired, or below a spec threshold).

1. Run `IMGLIFE-Report.ps1` to identify devices meeting the criteria (enrollment/age and hardware fields).

1. Add those devices to `Device Lifecycle – Aging – Refresh-Due` and procure their replacements.

Step 7 — Implement secure retirement and wipe

1. For a device at end of life, add it to `Device Lifecycle – End-of-Life – Retire`.

2. Preview the action first: run `IMGLIFE-Remediation.ps1` (WhatIf by default) to confirm the target device and the chosen action.

1. In Intune, Devices → All devices → (the device) → choose Retire (remove company data/management) or Wipe (full factory reset) per your documented policy. Wipe is the secure full reset; Retire keeps the OS but removes company data and management.

1. After confirming the wipe completed, remove the device record and de-register it from Autopilot (Devices → Enrollment → Windows Autopilot → Devices → Delete) so inventory stays clean.

Step 8 — Pilot the full flow

1. Run the whole procure → deploy → refresh → retire flow on the `Autopilot Provisioning – Standard Users – Pilot` group.
1. Validate with `IMGLIFE-Validation.ps1` (provisioning/enrollment) and `IMGLIFE-Report.ps1` (inventory and reconciliation). Complete the Testing Guide.

Step 9 — Expand to production

1. After the pilot passes, assign the broad Autopilot profile and onboard the rest of the fleet.
2. Run the lifecycle on the broad population; re-reconcile inventory each cycle with `IMGLIFE-Report.ps1`.

Order of operations

```
Lifecycle groups → register devices + Autopilot profile (or ConfigMgr imaging) → establish inventory → Deployed stage → refresh criteria → secure retire/wipe → pilot → expand
```

5. Testing Guide — Endpoint Imaging, Inventory & Hardware Lifecycle Management

Run after the pilot flow applies (Implementation Step 8) and each lifecycle cycle.

Test 1 — Provisioning method works

Do: Provision a pilot device by the chosen method — for Autopilot, reset/boot a registered device and complete OOBE; for imaging, run the task sequence. Then run `IMGLIFE-Validation.ps1`. Expected: The device provisions to the standard state and enrolls in Intune.

Test 2 — Autopilot registration present (if Autopilot)

Do: Intune → Devices → Enrollment → Windows Autopilot → Devices; or run `IMGLIFE-Validation.ps1`. Expected: The pilot device's hardware identity is registered and assigned the deployment profile.

Test 3 — Device enrolled and reporting hardware

Do: Intune → Devices → All devices → (the device) → Hardware; or run `IMGLIFE-Validation.ps1`. Expected: The device is present and reports model, serial number, storage, and memory.

Test 4 — Inventory is the source of truth

Do: Run `IMGLIFE-Report.ps1`. Expected: The fleet inventory lists the device with its hardware details and lifecycle stage.

Test 5 — Reconciliation finds gaps

Do: Compare the procurement list (serials) against `IMGLIFE-Report.ps1` output. Expected: Procured-but-not-deployed and deployed-but-not-inventoried devices are flagged.

Test 6 — Refresh-due detection

Do: Apply the refresh criteria and run `IMGLIFE-Report.ps1`. Expected: Devices meeting the criteria appear and are eligible for `Device Lifecycle – Aging – Refresh-Due`.

Test 7 — Secure retire/wipe (controlled)

Do: On a disposable pilot device, preview with `IMGLIFE-Remediation.ps1` (WhatIf), then perform Retire or Wipe in Intune and confirm completion. Expected: The device returns to a clean state (Wipe) or loses company data/management (Retire), then is removed from inventory and de-registered from Autopilot.

Pass/fail gate

Gate	Requirement
Pilot pass	Tests 1–5 pass on the pilot group; a test retire/wipe (Test 7) completes cleanly
Broad pass	Tests 3–6 pass across the broad population before the next refresh cycle

6. Troubleshooting Guide — Endpoint Imaging, Inventory & Hardware Lifecycle Management

Use `IMGLIFE-Validation.ps1` / `IMGLIFE-Discovery.ps1` for device state, `IMGLIFE-Report.ps1` for fleet inventory and reconciliation, and `IMGLIFE-Remediation.ps1` for a safe retire/wipe (preview by default).

A. Autopilot provisioning problems

Symptom	Cause	Remediation
Device skips Autopilot at OOBE	Hardware identity not registered, or no profile assigned	Confirm the device is in Devices → Enrollment → Windows Autopilot → Devices; assign the deployment profile to a group the device is in; run `IMGLIFE-Validation.ps1`

Symptom	Cause	Remediation
Profile assigned but not applied	Profile not yet synced to the device record	Wait for assignment to show Assigned; reset the device and retry OOBE
Wrong join type / OOBE options	Profile settings differ from intent	Edit the Autopilot deployment profile (join type, OOBE) and reassign

B. Imaging (Configuration Manager) problems

Symptom	Cause	Remediation
Task sequence fails / device won't PXE	Boot media, distribution point, or content missing	Confirm the OS image and task sequence are distributed to the distribution point and the device can reach PXE/boot media
Imaged device not in Intune	Enrollment step skipped after imaging	Ensure the task sequence (or post-image step) Entra-joins and auto-enrolls the device

C. Device enrolled but not reporting hardware

• Confirm the device is in Devices → All devices and recently checked in; force a sync from the device (Settings → Accounts → Access work or school → Info → Sync), then re-run `IMGLIFE-Validation.ps1`.

D. Inventory and reconciliation gaps

Symptom	Cause	Remediation
Purchased device not in inventory	Bought but never enrolled	Provision/enroll the device, or mark it `Device Lifecycle – Procurement – Procured`; re-run `IMGLIFE-Report.ps1`
Device in Intune but not on the books	Procurement list incomplete	Add the serial to the procurement record; reconcile again
Stale/duplicate records	Retired device not removed	Remove the device record and de-register from Autopilot after wipe

E. Retire / wipe issues

• Wipe pending forever: the device must be online to receive the command; confirm it checks in. If it will never return, after the documented hold period remove the record so inventory stays clean.

• Wrong device almost wiped: always run `IMGLIFE-Remediation.ps1` in its default WhatIf mode first

and confirm the serial/name before applying.

F. Diagnostics quick reference

```
Get-ComputerInfo | Select-Object CsManufacturer, CsModel, BiosSeralNumber
dsregcmd /status        # Entra join + MDM enrollment state
Get-Service -Name dmwappushservice, IntuneManagementExtension -ErrorAction SilentlyContinue
```

For the cloud-side device, Autopilot, and inventory state, use `IMGLIFE-Discovery.ps1` and `IMGLIFE-Report.ps1` (Microsoft Graph).

G. When to escalate

If a device is registered to Autopilot, assigned a profile, online, and synced, yet still fails to provision or never reports hardware after a full reset and sync, collect the device's enrollment diagnostics (Settings → Accounts → Access work or school → Export) and open a Microsoft support case.

7. Best Practices — Endpoint Imaging, Inventory & Hardware Lifecycle Management

• Choose one provisioning method deliberately. Default to Windows Autopilot cloud provisioning; keep Configuration Manager task-sequence imaging only for sites/device types that genuinely need a custom on-disk image or offline deployment. Don't run both for the same population by accident.

• Register hardware identity at procurement. Capture the hardware hash (or have the OEM/reseller register devices to Autopilot) before the device reaches the user, so deployment is zero-touch.

• Make Intune the single source of truth. Reconcile against the enrolled-device inventory, never a side spreadsheet; a spreadsheet drifts the moment it is saved.

• Reconcile every cycle. Compare purchased serials against `IMGLIFE-Report.ps1` to catch procured-but-not-deployed and deployed-but-not-inventoried devices early.

• Use lifecycle-stage groups, named consistently. Move devices through `Device Lifecycle – Procurement – Procured`, `– Fleet – Deployed`, `– Aging – Refresh-Due`, and `– End-of-Life – Retire`, and reuse those exact names everywhere.

• Define refresh criteria up front. Decide the trigger (age, warranty, spec) so refresh is a rule, not a guess, and procure replacements before devices fail.

• Always preview a wipe. Run `IMGLIFE-Remediation.ps1` in its default WhatIf mode and confirm the target device before any retire/wipe — the action is irreversible.

• Retire clean, every time. Choose Retire vs. Wipe deliberately, confirm completion, then remove the record and de-register from Autopilot so inventory stays accurate.

• Pilot before the fleet. Validate the full procure → deploy → refresh → retire flow on the pilot

group before expanding.

• Document the lifecycle. Record the chosen provisioning method, refresh criteria, and the retire/wipe authorization for audits and repeatability.

8. Deliverables Summary — Endpoint Imaging, Inventory & Hardware Lifecycle Management

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	The lifecycle model, the Autopilot-vs-imaging decision, inventory as source of truth, the procure/deploy/refresh/retire flow with secure wipe
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Infrastructure)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered steps for provisioning, inventory, lifecycle stages, and secure retirement
5. Testing Guide	Validation procedures with pass/fail gates
6. Troubleshooting Guide	Field issues and remediation
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`IMGLIFE-Discovery.ps1`	Reports current provisioning state: Autopilot registration, Intune enrollment, hardware inventory, and any ConfigMgr client (read-only)
`IMGLIFE-Validation.ps1`	Confirms a device is provisioned, registered (Autopilot), enrolled, and reporting hardware (read-only)
`IMGLIFE-Report.ps1`	Lifecycle and asset-reconciliation report across managed devices and the Autopilot registered-devices list (Microsoft Graph, read-only)
`IMGLIFE-Remediation.ps1`	Safe retire/wipe helper; previews by default (WhatIf), triggers the chosen end-of-life action only when explicitly applied
`IMGLIFE-Set-LifecycleStageTargeting.ps1`	Creates the lifecycle-stage Entra ID groups (idempotent) and optionally stages members

What the buyer receives

- This implementation guide as a branded Word document and PDF.
- All five PowerShell scripts in a single ZIP.
- A repeatable process following the model: Assess → Configure → Pilot → Validate → Expand.

Outcome

Every device accounted for across its whole life — provisioned by a defined method, inventoried accurately, refreshed on criteria, and retired with a secure wipe.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

IMGLIFE-Discovery.ps1

Purpose: Inventories the local device's lifecycle-relevant state so you know how it is provisioned and managed: OS build and hardware identity (manufacturer/model/serial), Entra join + Intune (MDM) enrollment via dsregcmd, the Intune Management Extension service, and whether a Configuration Manager (SCCM) client is present (the imaging alternative). Run on a device to confirm where it sits in the procure -> deploy -> refresh -> retire flow. HOW TO RUN Elevated, on the device: powershell -ExecutionPolicy Bypass -File .\IMGLIFE-Discovery.ps1 EXPECTED RESULTS A summary of OS, hardware identity, Entra/MDM enrollment state, Intune extension presence, and whether the ConfigMgr client is installed - so you can tell Autopilot/Intune-managed devices from task-sequence-imaged (ConfigMgr) ones. .NOTES Infrastructure solution - the repeatable way. Read-only (no changes made). #> [CmdletBinding()] param()

How to run: Elevated, on the device: powershell -ExecutionPolicy Bypass -File .\IMGLIFE-Discovery.ps1 EXPECTED RESULTS A summary of OS, hardware identity, Entra/MDM enrollment state, Intune extension presence, and whether the ConfigMgr client is installed - so you can tell Autopilot/Intune-managed devices from task-sequence-imaged (ConfigMgr) ones. .NOTES Infrastructure solution - the repeatable way. Read-only (no changes made). #> [CmdletBinding()] param()

IMGLIFE-Remediation.ps1

Purpose: Performs the secure end-of-life action on a managed device via Microsoft Graph: either Retire (remove company data and management, leave the OS) or Wipe (full factory reset - the secure wipe). It looks the device up by serial number, shows exactly what it WOULD do, and - because a wipe is irreversible - defaults to -WhatIf so it makes NO changes until you pass -WhatIf:$false. Optionally de-registers the device from Windows Autopilot afterwards so inventory stays clean. HOW TO RUN Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\IMGLIFE-Remediation.ps1 ` -SerialNumber "" -Action Wipe Apply the action: ... -SerialNumber "" -Action Wipe -WhatIf:$false Also de-register from Autopilot after the action: ... -SerialNumber "" -Action Wipe -RemoveAutopilot -WhatIf:$false Requires the Microsoft Graph PowerShell SDK and consent for DeviceManagementManagedDevices.PrivilegedOperations.All and DeviceManagementManagedDevices.ReadWrite.All (and DeviceManagementServiceConfig.ReadWrite.All for -RemoveAutopilot), all delegated. Sign in when prompted. EXPECTED RESULTS With -WhatIf (default): a plan of the retire/wipe (and optional Autopilot removal) that would occur, with the matched device's name/model for confirmation. With -WhatIf:$false: the chosen action is issued to the device and, if requested, the device is de-registered from Autopilot. .NOTES Infrastructure solution - the repeatable way. Default -WhatIf; irreversible action only when disabled. #> [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')] param([Parameter(Mandatory = $true)][string]$SerialNumber, [Parameter(Mandatory = $true)][ValidateSet('Retire', 'Wipe')][string]$Action, [switch]$RemoveAutopilot)

How to run: Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\IMGLIFE-Remediation.ps1 ` -SerialNumber "<device-serial>" -Action Wipe Apply the action: ... -SerialNumber "<device-serial>" -Action Wipe -WhatIf:$false Also de-register from Autopilot after the action: ... -SerialNumber "<serial>" -Action Wipe -RemoveAutopilot -WhatIf:$false Requires the Microsoft

Graph PowerShell SDK and consent for DeviceManagementManagedDevices.PrivilegedOperations.All and DeviceManagementManagedDevices.ReadWrite.All (and DeviceManagementServiceConfig.ReadWrite.All for -RemoveAutopilot), all delegated. Sign in when prompted. EXPECTED RESULTS With -WhatIf (default): a plan of the retire/wipe (and optional Autopilot removal) that would occur, with the matched device's name/model for confirmation. With -WhatIf:$false: the chosen action is issued to the device and, if requested, the device is de-registered from Autopilot. .NOTES Infrastructure solution - the repeatable way. Default -WhatIf; irreversible action only when disabled. #> [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')] param([Parameter(Mandatory = $true)][string]$SerialNumber, [Parameter(Mandatory = $true)][ValidateSet('Retire', 'Wipe')][string]$Action, [switch]$RemoveAutopilot)

IMGLIFE-Report.ps1

Purpose: Produces the inventory source-of-truth report across two Microsoft Graph surfaces: (1) Intune managed devices (manufacturer, model, serial, storage, enrollment + last-sync dates), and (2) the Windows Autopilot registered-devices list (the procurement-to-deployment ledger). It then reconciles them so gaps are obvious: devices registered to Autopilot but not enrolled in Intune (procured/registered but not deployed), and enrolled devices flagged as refresh-due by an age threshold. This is the report you reconcile against your purchase list. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\IMGLIFE-Report.ps1 -RefreshAfterDays 1460 -RefreshAfterDays sets the age (since enrollment) at which a device is flagged refresh-due (default 1460 = ~4 years). Requires the Microsoft Graph PowerShell SDK and consent for DeviceManagementManagedDevices.Read.All and DeviceManagementServiceConfig.Read.All (delegated). EXPECTED RESULTS A managed-device inventory table, the Autopilot registered list, a reconciliation of registered-but-not-enrolled devices, and a refresh-due list - so coverage and aging gaps are clear before you procure, refresh, or retire. .NOTES Infrastructure solution - the repeatable way. Read-only (GET only against Graph). #> [CmdletBinding()] param([int]$RefreshAfterDays = 1460)

How to run: powershell -ExecutionPolicy Bypass -File .\IMGLIFE-Report.ps1 -RefreshAfterDays 1460 -RefreshAfterDays sets the age (since enrollment) at which a device is flagged refresh-due (default 1460 = ~4 years). Requires the Microsoft Graph PowerShell SDK and consent for DeviceManagementManagedDevices.Read.All and DeviceManagementServiceConfig.Read.All (delegated). EXPECTED RESULTS A managed-device inventory table, the Autopilot registered list, a reconciliation of registered-but-not-enrolled devices, and a refresh-due list - so coverage and aging gaps are clear before you procure, refresh, or retire. .NOTES Infrastructure solution - the repeatable way. Read-only (GET only against Graph). #> [CmdletBinding()] param([int]$RefreshAfterDays = 1460)

IMGLIFE-Set-LifecycleStageTargeting.ps1

Purpose: Creates the Entra ID security groups used to target each stage of the device lifecycle, using the solution naming convention, and optionally stages initial members in the Deployed group. These groups are then used for policy targeting and lifecycle reporting as devices move from procured to deployed to refresh-due to retired. Default object names (created if missing): Device Lifecycle - Procurement - Procured Device Lifecycle - Fleet - Deployed Device Lifecycle - Aging - Refresh-Due Device Lifecycle - End-of-Life - Retire HOW TO RUN Create all four groups (idempotent - reuses any that already exist): powershell -ExecutionPolicy Bypass -File .\IMGLIFE-Set-LifecycleStageTargeting.ps1 Create and add members to the Deployed group: ... -DeployedMemberIds @('','') Requires the Microsoft Graph PowerShell SDK and consent for Group.ReadWrite.All (delegated). EXPECTED RESULTS The four lifecycle groups exist (created or found) and any supplied Deployed members are added; the script prints each group's

object id for use in policy assignment and reporting. .NOTES Infrastructure solution - the repeatable way. Creates groups only; does not move or wipe devices. #> [CmdletBinding(SupportsShouldProcess = $true)] param([string]$ProcuredGroupName = 'Device Lifecycle - Procurement - Procured', [string]$DeployedGroupName = 'Device Lifecycle - Fleet - Deployed', [string]$RefreshDueGroupName = 'Device Lifecycle - Aging - Refresh-Due', [string]$RetireGroupName = 'Device Lifecycle - End-of-Life - Retire', [string[]]$DeployedMemberIds = @())

How to run: Create all four groups (idempotent - reuses any that already exist): powershell -ExecutionPolicy Bypass -File .\IMGLIFE-Set-LifecycleStageTargeting.ps1 Create and add members to the Deployed group: ... -DeployedMemberIds @('<entra-object-id-1>','<entra-object-id-2>') Requires the Microsoft Graph PowerShell SDK and consent for Group.ReadWrite.All (delegated). EXPECTED RESULTS The four lifecycle groups exist (created or found) and any supplied Deployed members are added; the script prints each group's object id for use in policy assignment and reporting. .NOTES Infrastructure solution - the repeatable way. Creates groups only; does not move or wipe devices. #> [CmdletBinding(SupportsShouldProcess = $true)] param([string]$ProcuredGroupName = 'Device Lifecycle - Procurement - Procured', [string]$DeployedGroupName = 'Device Lifecycle - Fleet - Deployed', [string]$RefreshDueGroupName = 'Device Lifecycle - Aging - Refresh-Due', [string]$RetireGroupName = 'Device Lifecycle - End-of-Life - Retire', [string[]]$DeployedMemberIds = @())

IMGLIFE-Validation.ps1

Purpose: Validates that a device is correctly through the deploy stage of the lifecycle: locally it confirms Entra join + MDM enrollment (dsregcmd) and that hardware identity is present; via Microsoft Graph it confirms the device exists in Intune managed devices (reporting hardware) and, if a serial is supplied, that it is registered to Windows Autopilot. PASS/WARN lines make readiness obvious before you move the device into the Deployed stage. HOW TO RUN Local checks only: powershell -ExecutionPolicy Bypass -File .\IMGLIFE-Validation.ps1 Include cloud checks (Intune + Autopilot) for a specific serial: ... -SerialNumber "" Cloud checks require the Microsoft Graph PowerShell SDK and consent for DeviceManagementManagedDevices.Read.All and DeviceManagementServiceConfig.Read.All (delegated). EXPECTED RESULTS PASS/WARN lines for Entra join, MDM enrollment, local hardware identity, Intune managed-device presence, and Autopilot registration (when a serial is supplied). .NOTES Infrastructure solution - the repeatable way. Read-only (GET only against Graph). #> [CmdletBinding()] param([string]$SerialNumber)

How to run: Local checks only: powershell -ExecutionPolicy Bypass -File .\IMGLIFE-Validation.ps1 Include cloud checks (Intune + Autopilot) for a specific serial: ... -SerialNumber "<device-serial>" Cloud checks require the Microsoft Graph PowerShell SDK and consent for DeviceManagementManagedDevices.Read.All and DeviceManagementServiceConfig.Read.All (delegated). EXPECTED RESULTS PASS/WARN lines for Entra join, MDM enrollment, local hardware identity, Intune managed-device presence, and Autopilot registration (when a serial is supplied). .NOTES Infrastructure solution - the repeatable way. Read-only (GET only against Graph). #> [CmdletBinding()] param([string]$SerialNumber)